

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	
Project Title	HDI Predictor using Machine Learning and Flask
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This project aims to create a user-friendly HDI Predictor platform to automate the analysis of human development metrics. By leveraging machine learning and web technologies, the solution provides researchers and policy analysts with rapid, accurate predictions and clear data visualizations.

Project Overview	
Objective	Build a Python Flask web application that uses a Scikit-Learn Linear Regression model (serialized as HDI.pkl) to predict the UN Human Development Index (HDI) based on life expectancy, expected years of schooling, mean years of schooling, and GNI per capita.
Scope	Designed to assist researchers, students, and policy analysts by providing rapid, automated, and visually intuitive HDI data analysis.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction.
Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system.
Key Features	- Single predictions for HDI scores.

	- Batch CSV uploads for bulk analysis. - Country auto-fill using local CSV data. - Interactive visualizations using Chart.js.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv